

What is Cloud DevOps?

Definition

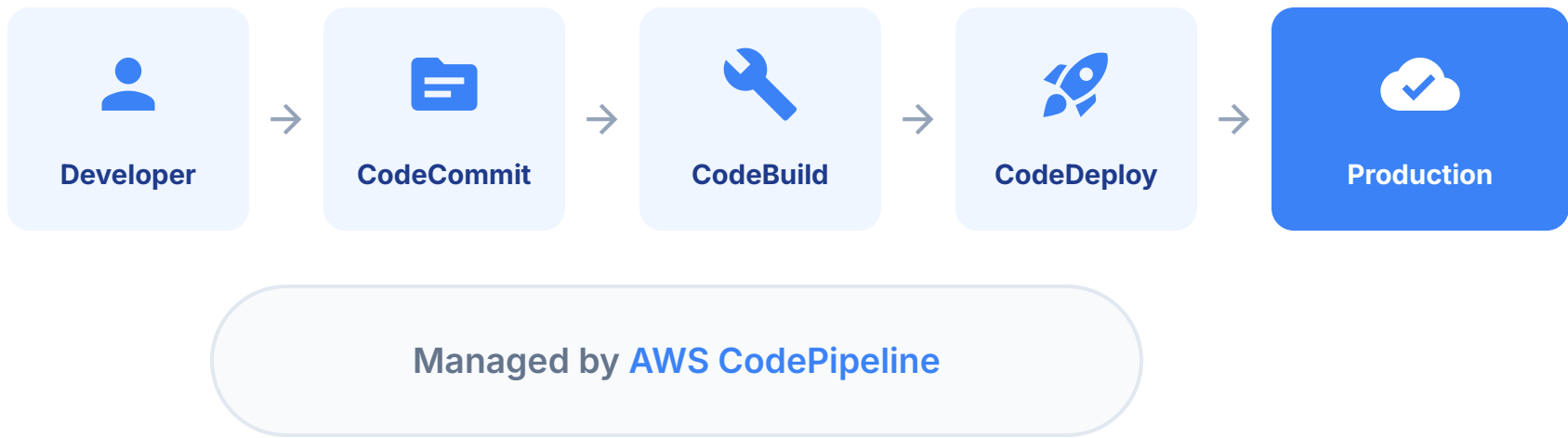
Cloud DevOps is the powerful combination of three core pillars:

- 01 Development (Dev)
- 02 Operations (Ops)
- 03 Cloud Services

Key Benefits

-  Faster deployments
-  Improved scalability
-  Reduced manual effort
-  Better reliability
-  Continuous delivery

AWS CI/CD Pipeline Flow



Key Features & Benefits

Key Features

-  Fully managed services
-  Easy integration
-  Auto scaling
-  High availability
-  Security and compliance

Benefits

-  Faster software releases
-  Reduced operational overhead
-  Improved automation

Introduction to AWS CodeCommit

What is CodeCommit?

AWS-managed Git repository service.

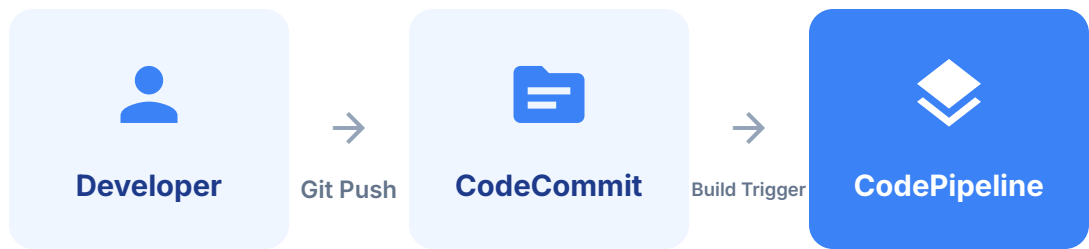
Key Features

-  Private Git repositories
-  Secure access & IAM integration
-  Encryption at rest
-  High availability

Benefits

-  No Git server management
-  Scalable infrastructure
-  Reliable & Enterprise-ready

CodeCommit Architecture



Managed architectural flow for [AWS Developer Tools](#)

CodeCommit Capabilities

Supported Operations

 Clone

 Commit

 Push

 Pull

 Branch

 Merge

Use Cases

 Application source code

 Infrastructure code (IaC)

 Configuration files

AWS CodeBuild Overview

What is CodeBuild?

A fully managed, continuous integration service that compiles source code, runs tests, and produces software packages.

Core Functions

-  Compile source code
-  Run unit tests
-  Create software packages
-  Build Docker images

Build Workflow

INPUT

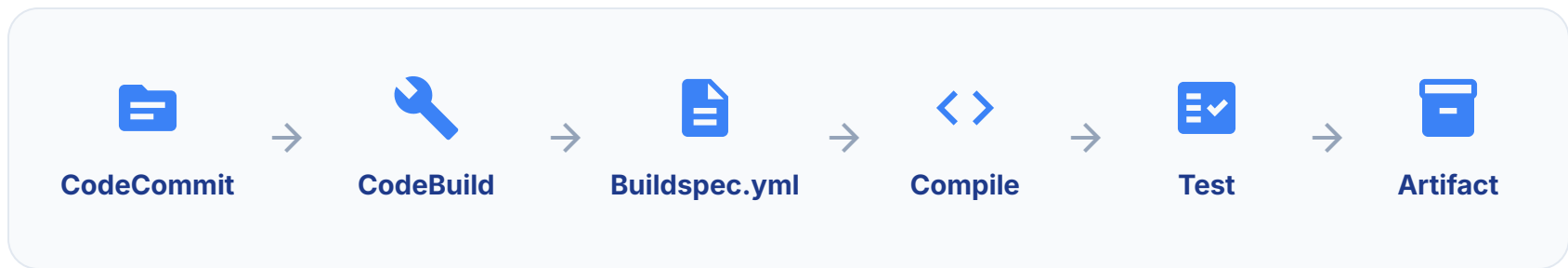
Source Code (GitHub, S3, CodeCommit)



OUTPUT

- Artifacts & Binary files
- Docker Container Images
- Detailed Build Reports

CodeBuild Workflow



Advantages

- ✓ Serverless builds: No need to provision or manage servers.
- ✓ Pay only for usage: Charged by the minute for build resources used.
- ✓ Parallel builds: Scale build capacity to handle concurrent builds.
- ✓ Easy integration: Works seamlessly with AWS services and popular tools.

Sample buildspec.yml

```
version: 0.2

phases:
  install:
    commands:
      - echo Installing
  build:
    commands:
      - echo Building
  post_build:
    commands:
      - echo Completed

artifacts:
  files:
    - '**/*'
```



Purpose

Defines build instructions for CodeBuild.

KEY COMPONENTS

- Build Phases
- CLI Commands
- Artifact Output

AWS CodeDeploy Overview

What is CodeDeploy?

Automates application deployments to various computing services.

Supported Platforms

-  Amazon EC2
-  On-Premises Servers
-  AWS Lambda
-  Amazon ECS

Key Benefits

- ✓ Automated deployments
- ✓ Reduced downtime
- ✓ Rollback support
- ✓ Deployment monitoring

CodeDeploy Deployment Strategies

In-Place Deployment

Old Version



Replace



New Version

Blue/Green Deployment

Blue Environment



Switch Traffic

Green Environment

Advantages of Blue/Green

- ✓ Near-zero downtime
- ✓ Easy rollback
- ✓ Safer releases

appspec.yml Example

```
version: 0.0
os: linux

files:
  - source: /
    destination: /var/www/html

hooks:
  AfterInstall:
    - location: scripts/start.sh
```



Purpose

Controls deployment behavior in CodeDeploy.

Key Components

- File Mapping
- Lifecycle Hooks
- OS Specification

AWS CodePipeline Overview

What is CodePipeline?

Continuous Integration and Continuous Delivery service.

Pipeline Stages

 Source



 Build



 Test



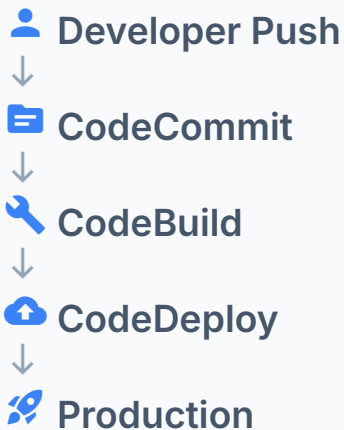
 Deploy

Supported Integrations

- ✓ CodeCommit
- ✓ GitHub
- ✓ CodeBuild
- ✓ CodeDeploy
- ✓ CloudFormation

CodePipeline End-to-End Flow

Pipeline Process



Benefits

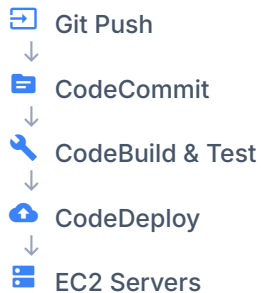
- ✓ End-to-end automation
- ✓ Faster releases
- ✓ Consistent deployments
- ✓ Reduced human errors

CI/CD Pipeline Example

Scenario

Node.js Application Deployment

Pipeline Process



Results

- ✓ Automated deployment
- ✓ Faster delivery
- ✓ Consistent environments